Distance sensor

DETECT | RASPBERRY PI | DISTANCE

What does it do?

An HC-SR04P distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

Uses



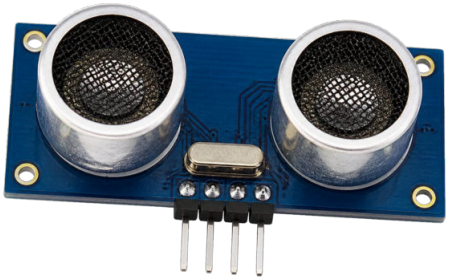
Parking sensor

DETECT: Gap to the wall → DECIDE: Is it closer than 30 cm? → DO: Beep faster



Air harp

DETECT: How far your hand is → DECIDE: Which note is that? → DO: Play the note



VCC	Trig	Echo	GND
Power, 3V3	Sends a click	Times the return	Ground

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import DistanceSensor

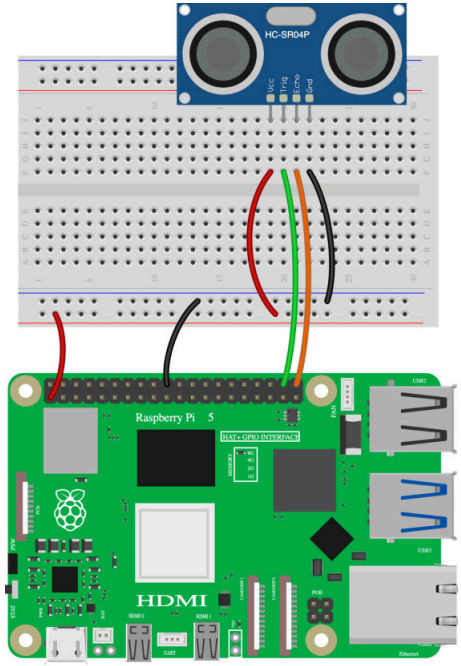
sensor = DistanceSensor(
    echo=21, trigger=20
)

distance = sensor.distance
```

`distance` is in metres and stops at 1. Pass `max_distance=2` to see further, or read `sensor.value` for 0 to 1. Use an **HC-SR04P**, powered from **3V3**.

Wiring

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HC-SR04P · **3V3** (pin 1) to + rail · Trig **GPIO 20** (pin 38) · Echo **GPIO 21** (pin 40) · **GND** (pin 20) to − rail